

List Of Data Analysis Projects

1. **MySQL + Power BI projects** (SQL-driven analytics, reporting, dashboards)
2. **Python + Power BI projects** (EDA, transformation, automation + visualization)

Each project reflects **real business problems**, uses **realistic datasets**, and is suitable for **portfolio, interviews, and classroom teaching**.

1. MySQL + Power BI Project Ideas

(Focus: SQL querying, data modeling, KPI creation, dashboarding)

1. Retail Sales Performance Analysis

Business Problem:

Management wants visibility into sales trends and underperforming regions.

Key Analysis:

- Sales by region, category, sub-category
- Monthly & quarterly growth
- Top vs bottom products
- Profit margin analysis

SQL Skills Used:

GROUP BY, JOIN, CTE, WINDOW FUNCTIONS, DATE FUNCTIONS

Power BI Output:

Sales dashboard with slicers (region, date, category)

2. Coffee Shop Sales & Operations Dashboard

Business Problem:

Optimize store operations and peak-hour staffing.

Key Analysis:

- Hourly sales trend

Created By: Aishwarya Mate

- Average order value
- Store-wise performance
- Best-selling products

Why Interview-Friendly:

Simple domain, strong KPIs, excellent for explaining logic.

3. Customer Churn Analysis (Telecom / Bank)

Business Problem:

Identify churn patterns and risk factors.

Key Analysis:

- Churn rate by tenure, age, geography
- Revenue lost due to churn
- Active vs inactive customers

SQL Focus:

Conditional aggregation, CASE WHEN, subqueries

4. Inventory & Supply Chain Analysis

Business Problem:

Reduce stockouts and excess inventory.

Key Analysis:

- Inventory turnover
 - Order fulfillment time
 - Supplier performance
 - Stock aging
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5. E-commerce Order & Delivery Analysis

Created By: Aishwarya Mate

Business Problem:

Improve delivery timelines and customer satisfaction.

Key Analysis:

- Order-to-delivery duration
 - Delayed orders by city
 - Payment mode trends
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6. HR Analytics Dashboard

Business Problem:

Monitor attrition and workforce productivity.

Key Analysis:

- Attrition rate
 - Experience vs salary analysis
 - Department-wise turnover
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7. Banking Loan Performance Analysis

Business Problem:

Assess loan risk and repayment patterns.

Key Analysis:

- Default rate by income group
 - Loan approval trends
 - Risk segmentation
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8. Hotel Booking & Revenue Analysis

Business Problem:

Increase occupancy and revenue.

Key Analysis:

Created By: Aishwarya Mate

- Booking cancellation rate
 - Seasonal demand
 - Revenue per room
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2. Python + Power BI Project Ideas

(Focus: EDA, data cleaning, automation, feature engineering)

1. Sales Data Exploratory Data Analysis & Dashboard

Python Work:

- Missing value handling
- Outlier detection
- Feature engineering (Year, Month, Weekday)

Power BI:

- Interactive trend dashboard
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2. Customer Segmentation using Python + Power BI

Python:

- Data Analysis in Python

Power BI:

- Segment-wise revenue dashboard
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3. Marketing Campaign Effectiveness Analysis

Python:

- Campaign ROI
- Conversion rate analysis

Power BI:

- Campaign comparison dashboard

4. E-commerce Product Recommendation Insights

Python:

- Product affinity analysis
- Frequently bought together

Power BI:

- Product cross-sell dashboard
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5. Data Pipeline Project

Python:

- Extract CSV → Clean → Load to MySQL
- Scheduled scripts

Power BI:

- Live dashboard connected to MySQL
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3. Best Projects for Freshers (Top 5 Recommendation)

If you want **maximum interview impact**, prioritize these:

1. Coffee Shop Sales (MySQL + Power BI)
 2. Retail Sales Performance (MySQL + Power BI)
 3. Customer Churn Analysis (MySQL + Power BI)
 4. Customer Segmentation (Python + Power BI)
 5. ETL Pipeline (Python + MySQL + Power BI)
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4. How Interviewers Evaluate These Projects

They look for:

- Clear **business problem statement**

Created By: Aishwarya Mate

- Correct **KPIs**
 - Clean **SQL logic**
 - Well-structured **Power BI dashboard**
 - Ability to **explain insights**, not just visuals
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Data Resources:

1. Kaggle (Primary Source)

Best for: Clean CSVs, realistic business datasets, no access issues.

Use cases:

- Retail sales
- Customer churn
- HR analytics
- E-commerce
- Banking
- Marketing

Typical datasets:

- Superstore Sales
- Telecom Churn
- E-commerce Orders
- Bank Customer Data
- HR Attrition

2. Data.gov (US Government Open Data)

Best for: Time series, demographics, logistics, transport, public services.

Use cases:

- Supply chain

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- **Transportation delays**
- **Employment trends**

3. Google Dataset Search

Best for: Discovering datasets across universities, companies, and portals.

Search keywords:

- **"Retail sales dataset CSV"**
- **"Customer churn dataset"**
- **"E-commerce orders data"**